

The use and abuse of Pb in bioarchaeological studies: A review of Pb concentration and isotope analyses of teeth

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ABSTRACT

Lead (Pb) concentration and isotope analyses in human remains are powerful analytical tools used to examine differences in Pb exposure over time or between populations and to examine the movement of peoples and Pb-containing cultural materials. While there was a large increase in the use of dental tissues for measuring Pb in the last 30 years, there has yet to be a critical evaluation of how these analyses are conducted or the data are used to answer archaeological questions. This article reviews 55 papers published between 1979 and 2021 on Pb concentrations and isotope analyses of teeth from archaeological populations to examine how they were used, areas where they were misused or insufficient information given, current limitations of the approaches, and future studies needed. This was applied across three broad topics: sample choice, concentration interpretation, and isotopic ratio interpretation.

While major limitations exist, largely related to missing information in methodological approaches, there are a few overarching themes of use that need to be considered. First is considering the biological/cultural age of the teeth and therefore of the individuals represented, and the variability introduced when comparing disparate tooth types. Second, is the need to consider archaeological, ethnographic, and historical documentation when evaluating natural versus anthropogenic exposures. Finally, we recommend that greater consideration of the contributions from both environmental and cultural sources, including those that could be imported into different cultural regions. The consideration of all these factors is integral to future studies involving Pb concentration and isotope analyses in bioarchaeology.

1. Introduction

Over the past 20 years, lead (Pb) has reemerged as an archaeological tool to improve our understanding of past human movement and the historical extent of pollution, either by the comparison of Pb concentrations (e.g., Budd et al., 2004; Schroeder et al., 2013) or by differences in the Pb isotope ratios of bioavailable Pb in the natural environment (e.g., Montgomery et al., 2000, 2005; Valentine et al., 2008; Fitch et al., 2012; Price et al., 2017). Initial studies examined Pb concentration and isotope ratios in bone (Jarcho 1964; Jaworowski 1968; Kowal et al., 1991), but tooth enamel allows researchers to access childhood exposure while limiting the potential of diagenetic alteration from the burial environment and has become the preferred tissue choice. Although there have been reviews on Pb use (Gulson 2008) and a meta-analysis of modern and archaeological Pb data (Kamenov and Gulson 2014), a

critical assessment of the state of the field is warranted. Environmental and anthropogenic Pb exposure is used in archaeology to examine health, movement, and social dynamics in past populations; however, aspects of the application have also been misused, which weakens the conclusions. This article reviews published research on Pb concentrations and isotope analyses of teeth from archaeological populations to identify current trends, limitations due to misuse or insufficient information, fundamental or knowledge-based limitations, and suggests future research. Together, these considerations will improve the Pb analyses' ability to provide insight into childhood exposures, movement, and exchange of cultural materials in the past.

Here, we first give a basic background of Pb research, Pb sources and uptake, and then present a critical review of studies found through a systematic query of research search engines and journals that frequently publish Pb data. The review focuses on how techniques are applied, and

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the approaches taken for interpretation. The first concern is sample selection and how it affects the interpretation of *in vivo* exposure, and the consideration of the child's experiences during the period of tooth development. The second concern is data interpretation and has two components: a) how the concentration in teeth can be meaningfully related to health issues, the timing of exposure, and population experiences, and b) the relationship between Pb isotope ratios and the characterization of both bioavailable and anthropogenic Pb sources.

2. Background to Pb studies

2.1. History of the use of Pb in bioarchaeology

The use of Pb concentration and isotope analyses in a bioarchaeological context developed in three main phases: measuring levels in bone, rising concerns about diagenetic influences, and a switch to using teeth. The first phase occurred in the 1960s to '70s when the initial implementation of Pb trace element analysis focused on bone as the analyte. Some of the main research themes from this period included examining what the archaeological 'natural' and anthropogenic exposure levels were compared to the modern period. Since atmospheric Pb pollution from leaded-gasoline and paint was a contemporary concern, researchers wanted to understand how modern exposures compared to those in past populations. Comparisons included populations with no known anthropogenic exposure (Jaworowski 1968; Grandjean and Holma 1973) and those with at least some anthropogenic Pb exposure (Jarcho 1964). There was also interest in Romano-British populations since Pb was widely used during that period (Mackie et al., 1975; Waldron et al., 1976). While studies in this first phase hypothesized that exposure was relatively low during periods with no known anthropogenic Pb sources, some studies did show that past exposures could be much higher than those in the modern period (e.g., Jaworowski 1968; Grandjean and Holma 1973). However, the Pb levels detected were accepted as accurate, and they did not question if the interpretation of exposures was affected by contamination in the burial environment.

The second development phase occurred in the 1980s and '90s when researchers began recognizing the effects of diagenesis on bone Pb concentrations. Waldron et al. (1979) first questioned the potential for diagenesis to impact the Pb content in bones since they found no correlation between soil Pb concentrations and bone concentrations, suggesting that the values were biogenic. In subsequent research, Waldron (1981, 1983) tested other sites and found that Pb uptake from the soil into bone occurred in some situations and potentially impacted the interpretation of *in vivo* Pb exposures. Studies continued to use bone Pb concentrations but utilized revised methods that were believed to remove the bone's diagenetic material (Aufderheide et al., 1985; Jaworowski et al., 1985; Patterson et al., 1987). However, the success of these methods could be variable, and studies concluded that bone was unlikely to retain biogenic values, even when treated to remove diagenetic material (Jaworowski et al., 1985; Patterson et al., 1991). This revelation resulted in a drop in the number of bioarchaeological studies that examined Pb in bulk samples of bone.

This second period was also when researchers began to apply these data and methods to examine social differences between populations based on the levels of Pb exposure (Corruccini et al., 1987; Aufderheide et al., 1988). Though it was not widespread, studies incorporated methods such as bone Pb isotope analysis (Kowal et al., 1991; Carlson 1996) and examined Pb concentrations of teeth (Ericson et al., 1979; Whittaker and Stack 1984; Grandjean and Jørgenson 1990; Patterson et al., 1991).

The most recent Pb research phase began in the early 2000s when bioarchaeological research increased tooth enamel use to look at past exposure levels (Fig. 1). In parallel, the debate of whether biogenic values of Pb isotope or trace element concentrations are accessible through bone continued and remains a contentious issue, with many arguing that biogenic values are not accessible, as is the case for

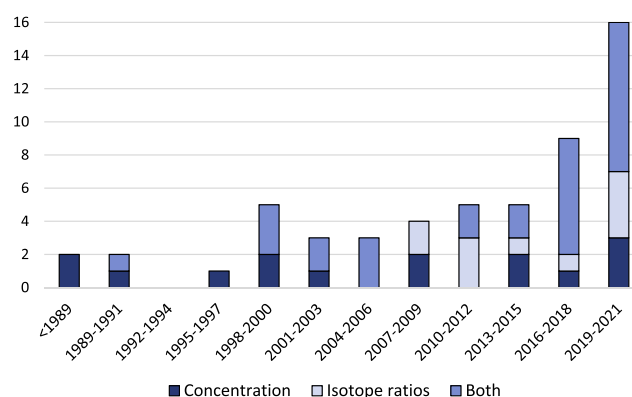


Fig. 1. Number of studies using Pb concentration and/or isotope ratios of archaeological teeth per year. Note the increase in studies since 1997 and the start of publications using only Pb isotope ratios in 2008.

strontium (Sr) in bone (Millard 2006; Zapata et al., 2006; Wittmers et al., 2008). However, this is not universally accepted, and bone tissue use continues with limited consideration of diagenesis in some cases (Bower et al., 2007; Rasmussen et al., 2015; Rasmussen et al., 2019; López-Costas et al., 2020). Many papers since 2011, when there was a resurgence of bone Pb studies, are exploring means of accessing biogenic Pb and the role of diagenesis in affecting these values (Swanston et al. 2012, 2018; McMillan et al., 2019; Simpson et al., 2021).

While there was a substantial increase in studies that exclusively used tooth enamel during this phase (e.g., Budd et al., 2000a, 2004; Montgomery et al., 2005, 2010; Valentine et al., 2008, 2015), there was also an increase in research using both bone and enamel, particularly in association with Sr isotope analysis to examine migration (Åberg et al., 1998; Budd et al., 2000a; Montgomery et al., 2000). The recent analysis themes include examining changes in Pb exposure and sources of exposure over time, similar to previous approaches involving bone but using teeth as an analyte (Budd et al. 2000a, 2004). There is also a consideration of childhood health using enamel Pb concentrations as a proxy for blood Pb concentrations (Schroeder et al., 2013; Millard et al., 2014; Jones 2015; Quinn et al., 2020; Moore et al., 2021). Studies are also considering migration, both through differences in Pb concentrations between anthropogenic and natural exposure regions (Schroeder et al., 2013; Jones 2015) and by examining differences in Pb isotope ratios (Montgomery et al., 2005; Fitch et al., 2012; Millard et al., 2014; Shaw et al., 2016; Laffoon et al., 2020).

Cultural affiliation is an important area of interest, with studies examining the 'cultural focusing' of Pb isotope ratios that result from an averaging of Pb sources within a population (Montgomery et al., 2005; Millard et al., 2014; Shaw et al., 2016). Some have used this to examine life histories of individuals' health and movement (e.g., Valentine et al., 2008; Fitch et al., 2012; Price et al., 2017; Laffoon et al., 2020). More recently, the stability or inertness of enamel has been questioned, suggesting that diagenetic Pb may be more prevalent in tooth enamel than previously believed (e.g., Beherec et al., 2016; Kamenov et al., 2018; King et al., 2020; Simonetti et al., 2021). From this brief overview of the development of Pb analyses in bioarchaeology, it is clear that Pb is still utilized to address meaningful research questions. However, there are also inconsistencies in how the technique is applied that must be further explored to understand challenges affecting the interpretation of Pb concentration and isotope ratios from archaeological skeletal material.

2.2. Sources of Pb

To examine how human skeletal materials are used and misused to address bioarchaeological questions, it is essential first to consider the sources of Pb variation within humans. Even without anthropogenic

contamination, natural processes result in different Pb concentrations and isotope ratios within human tissues. Additionally, physiological stages and cultural influences will also affect the deposition of Pb in skeletal materials. Understanding these sources of variability allows for questions concerning such themes as movement and Pb toxicity to be investigated.

There are four stable isotopes of Pb typically measured in bioarchaeological analyses: ^{206}Pb , ^{207}Pb , and ^{208}Pb , which are radiogenic, and ^{204}Pb , which is primordial (Faure and Mensing 2005). The three radiogenic isotopes are the stable daughter isotopes of uranium (^{238}U to ^{206}Pb , ^{235}U to ^{207}Pb) and thorium (^{232}Th to ^{208}Pb) decay chains with different half-lives (Doe 1970; Faure and Mensing 2005). The isotopes have different natural abundances, with ^{204}Pb being the lowest at 1.48%, ^{206}Pb and ^{207}Pb are similar with 23.5% and 22.6%, respectively, and ^{208}Pb has the highest abundance at 52.3% (Adriano 1986; Faure and Mensing 2005). The Pb isotope half-lives, the varied original isotopic abundances in rock, the age since and any mixing during its formation create the variability of Pb isotope ratios of geological formations (Gulson 1986; Adriano 1986; Teutsch et al., 2001; Faure and Mensing 2005). From these formations, there is no measurable isotopic fractionation of Pb as the isotopic ratios move through the biosphere as bioavailable Pb (Rabinowitz and Wetherill 1972; Gulson 1986) incorporating in the skeletal material of humans where they act as a reservoir of exposure history.

Pb is contained within the mineral galena (PbS), which was a common Pb source for metal-using societies and was also exposed for the silver within galena deposits, in addition to other minerals in lower concentrations (Adriano 1986; Brännvall et al., 1999). In addition to the parental rock material, various external sources can add to the concentration of soil Pb, including the dry deposition of dust, wet deposition via rainwater, and the deposition of eroded soil through river run-off (Adriano 1986). Pb has low natural concentrations in water of aquatic systems such as rivers, rain, and ocean water (Schaule and Patterson 1981; Bacon and Bain 1995). The turnover time is brief for river and rainwater (Bacon and Bain 1995) but can be 80–100 years for deep oceans (Schaule and Patterson 1981). While all these sources could have disparate Pb isotope ratios, their mixture within the soil is likely to result in an averaged value (Samuelson and Potra 2020). The soil, plants, and animals that occur and live within an area constitute the biologically available Pb to humans when there is no anthropogenic exposure (Fig. 2).

Many cultures have processed and incorporated Pb into cultural materials. However, Montgomery et al. (2010) argued that higher exposure was not associated with a society's ability to mine; instead, the contamination of the food and drink consumed resulted in high Pb exposure levels. Mining and smelting of Pb result in dust and particulate air contamination, which settles onto surfaces and into the soil where it can be ingested or incorporated when food is grown (Fig. 2; Gulson et al., 2004). Pb contamination from mining is not as bioavailable as with smelting due to the much larger size of the particles ($>150\text{ }\mu\text{m}$) (Steele et al., 1990; Rieuwerts et al., 2000). In turn, within 2 km of smelters, bioavailable Pb tends to be much higher, particularly in comparison to mining (Mahaffey 1978; Rieuwerts et al., 2000). Since Pb ions that substitute within biological systems, the specific molecules of the particle, the matrix the Pb is encased in, the pH of soil or water it is deposited into, and their solubility also impact bioavailability (Hemphill et al., 1991; Ram et al., 2021). For example, particulates from wastewater released at these mining and smelting sites may have high particulates, but sulfates are less soluble than Pb oxides, so not all of the particulates would be bioavailable (Ghotbizadeh et al., 2022). The mined Pb was often further processed and incorporated into food and drink storage vessels as an alloy or as an oxide into the glaze of ceramics (e.g., Beagrie 1989; Tite et al., 1998). It could also be added to food or drink as a sweetener or preservative (Lessler 1988) or inadvertently consumed if Pb shot was not removed from a harvested animal before preparation (Lévesque et al., 2003; Knott et al., 2010).

2.3. Pb uptake by humans

These natural and anthropogenic Pb sources make their way into the biosphere, where the primary means of exposure to humans is through dust, food, or cultural materials. Initial Pb exposure can occur through ingestion, inhalation, and skin contact (Flora et al., 2006). The main pathway of human exposure to Pb is through ingestion since it is often found in natural or contaminated food and drink (Flora et al., 2006). Plants accumulate most of their Pb in the root system via the soil, while the leaves are only a substantial repository of Pb when atmospheric pollution is present (Elias et al., 1982; Adriano 1986; Sharma and Dubey 2005). Dust is an additional source for humans, which is consumed, particularly by children, via hand-to-mouth activities (Mushak 1991; Gulson et al., 2004). Particle size plays a large role in the bioavailability of ingested Pb, with those between $63\text{ }\mu\text{m}$ and $250\text{ }\mu\text{m}$ accounting for

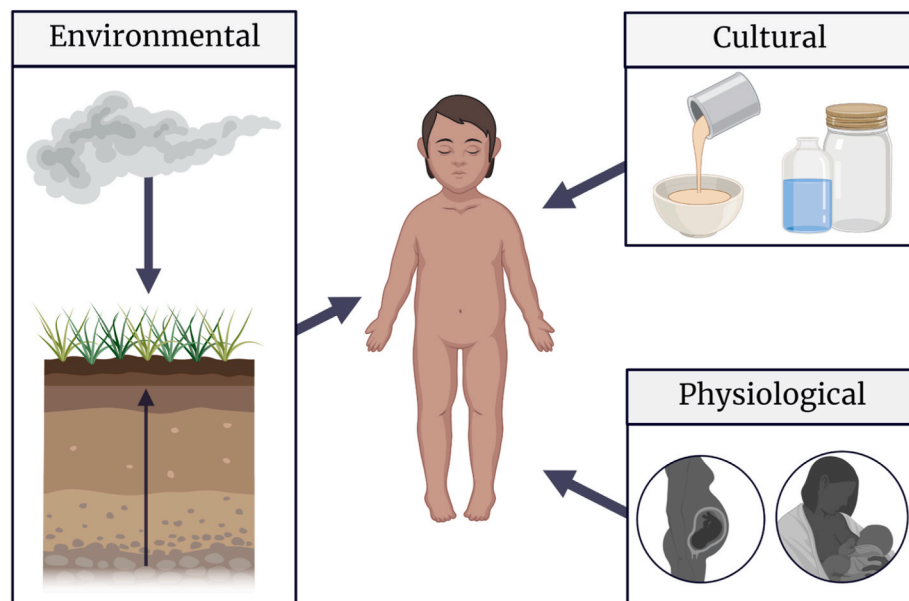


Fig. 2. The sources of geogenic and anthropogenic Pb to a human include environmental, cultural material, and physiological. The arrows depict the movement of Pb from the underlying bedrock into the soil and plants, but anthropogenic sources in the atmosphere can swamp the naturally occurring Pb. Food and drink that are stored in Pb-containing materials can leach the Pb resulting in further contamination. Finally, a mother or caregiver's previous Pb can be reabsorbed into the blood and released into a growing fetus or through breastmilk.

most of the exposure (Rieuwerts et al., 2000; Bright et al., 2006). Since the gastrointestinal tracts of children are not fully developed, they absorb and retain a higher proportion of Pb consumed with food (40%) when compared with adults (5–15%) (Mushak 1991; Flora et al., 2006). Additionally, dietary Pb is more effectively absorbed after fasting or with water (~100% in children and 35–60% in adults), so the medium of exposure impacts the amount of Pb absorbed (Rabinowitz et al., 1980:1788; ATSDR 2020). Conversely, the consumption of high concentrations of calcium, iron, magnesium, and phosphate can inhibit Pb absorption (Rabinowitz et al., 1980; Gulson et al., 2003).

Exposure via inhalation occurs from the suspended particulates in the air either from natural or anthropogenic sources, such as smelting emissions (Gulson et al., 2004; Flora et al., 2006). The amount of lead deposited into the respiratory tract varies, but for inorganic Pb it is approximately 25% (Morrow et al., 1980). Of that deposited into the bronchial or alveolar regions, particle size significantly impacts bioavailability. Particles >10 µm are easily expelled from the upper respiratory tract and particle size plays a large role in the bioavailability of inhaled Pb (Hornberg et al., 1998). However, those >2.5 µm are likely expelled and moved out of the bronchia with the mucosiliary escalator and, if not fully expelled, are consumed into the gastrointestinal tract (Oberdörster 1993; James et al., 1994; ATSDR 2020). While the absorption of Pb deposited into lung alveoli is more effective than the gut (90–95% versus 5–40%, respectively; Hursh et al., 1969; Wells et al., 1975; Flora et al., 2006:166), the concentration of atmospheric Pb is much lower than in food resulting in less exposure via air. It is estimated that 1–3% of Pb in humans comes from inhalation of atmospheric sources (Thornton et al., 1990) and that overall inhalation is a less like exposure pathway for children than occupationally exposed adults (WHO 2010; Zhou et al., 2014).

The final pathway of Pb absorption is direct skin exposure. This contribution is typically minimal to an individual's overall Pb burden since the skin does not readily absorb unpowdered Pb (Lilley et al., 1988; Filon et al., 2006; Flora et al., 2006; Pan et al., 2010). However, this likely had a considerable impact on those who practiced body painting, or those working with powdered Pb, particularly those in pottery manufacturing or Pb mining. One example is kohl. While kohl recipes vary and were not always lead-based (Riesmeier et al., 2022), modern studies did find that Egyptian eye makeup from as early as the 19th-century BCE contained Pb (Tapsoba et al., 2010; Riesmeier et al., 2022). Kohl's use would have resulted in some Pb exposure as is shown in modern studies (Al-Ashban et al., 2004). While not a concern for archaeological populations, organic Pb, or tetraethyl Pb, was widely used as a fuel additive from the 1920s until it was phased out in many countries by the mid-1990s and is easily absorbed by the skin (Needleman 2004).

While tactile, inhalation, and ingestion are initial means of exposure, mothers and caregivers can pass on their previous Pb exposure to fetuses and infants through maternal transfer before birth or via human milk. If mothers or caregivers have insufficient calcium (Ca) in their diet during pregnancy and lactation, bone turnover rates increase, releasing stored Ca and Pb (Gulson et al., 2003). Pb is then released into the bloodstream and can be passed onto the developing fetus or breastfeeding infant. Considering that deciduous teeth form *in utero* and during early infancy and that some permanent teeth develop while the infant may still be breastfeeding, the source of the Pb in the first teeth formed originates from the mother and/or a caregiver, and potentially remobilized Pb from those individuals (Gulson et al., 2003).

Once incorporated into the body, Pb is absorbed and distributed into three main systems (Rabinowitz et al., 1976). While blood contains a small portion of Pb compared to total body burden, it has a relatively fast turnover rate (36 ± 5 days), making it crucial for transporting Pb throughout the body (Rabinowitz et al., 1976). When Pb exposure is low, it is present in the red blood cells, but if high exposure occurs, plasma contains the majority of blood Pb (O'Flaherty, 1998; Hernández-Avila et al., 1998; Fleming et al., 1999). Pb is also incorporated into

soft tissues with a slightly longer turnover rate than blood and has been reported to be between 30 and 55 days (Rabinowitz et al., 1976). Pb is excreted from the body through feces, urine, and sweat (Rabinowitz et al., 1976; Flora et al., 2006). While soft tissue and blood are important for the absorption of Pb into humans, they do not often survive unless in unique archaeological contexts (O'Connell and Hedges 1999; Sharp et al., 2003; Finucane 2007).

Unlike blood and soft tissue, Pb has a prolonged turnover rate in bone, resulting in greater accumulation. Pb is also deposited within teeth; however, Pb in dental tissues makes up a much smaller portion of the skeletal system. There remain uncertainties with how Pb is incorporated into bioapatite, the mineral portion of bone and teeth. While many researchers suggest a simple substitution of Pb at Ca sites in hydroxyapatite (i.e., Verbeeck et al., 1981), Montgomery et al. (2010:211) pointed out that this ignores the chemical dissimilarities and original research by Neuman and Neuman (1958:94–95) who argued that it was not straightforward. Pb is also not uniformly distributed within teeth, with the outer 30 µm of enamel having a significantly higher concentration of Pb than the core enamel (Budd et al., 1998). This spike in concentration is present from the tooth's formation since unerupted and modern teeth have similar patterns, though some post-eruption addition is still possible (Budd et al., 1998). Dentine has about double the Pb concentration of the core enamel in both modern and archaeological teeth (Budd et al., 1998).

The use of archaeological teeth to examine Pb exposure has increased over the last 20 years. Multiple factors make this tissue a valuable resource to researchers. First, teeth form at various phases of physiological childhood, allowing the examination of multiple periods of exposure (Gulson and Wilson 1994; AlQahtani et al., 2010; see Table 1). Additionally, once the enamel forms and matures, the tissue does not remodel, meaning that the concentration and sources of Pb can be used to examine exposure during specific periods of life (Budd et al., 2000a; Hillson 2014). Unlike bone, enamel is mainly bioapatite with very little organic content (Hillson 2014). Enamel has very small pore spaces and the mineral is generally assumed to be highly stable, which would minimize the likelihood of contamination or diagenetic alteration. While this is repeated in Pb research, it should be noted that the foundational studies are not in the same context, with Budd et al. (2000b) examining Sr and Sponheimer and Lee-Thorp (2006) using fossilized teeth. Though, some studies suggest that enamel may be more prone to diagenesis than previously assumed (Beherec et al., 2016; King et al., 2020), particularly in cases where undermineralized enamel is used (Montgomery 2002). When properly sampled and vetted for diagenetic alteration, however, the Pb in biological tissues holds a wealth of information about past exposures.

In cases of 'natural' exposure, where minimal or no anthropogenic Pb exposure occurred, humans' Pb isotope ratios will represent an average of the sources consumed (Montgomery et al., 2005:129). This is not the case with anthropogenic Pb. It is essential to consider that there are typically a small number of anthropogenic Pb sources used by a culture

Table 1

Timing of the start of tooth formation, crown completion, and eruption of permanent teeth. Based on AlQahtani et al., (2010). AJPA 142:481–490.

Tooth type	Initial cusp formation		Crown completed with defined pulp roof		Occlusal surface in plane	
	maxilla	mandible	maxilla	mandible	maxilla	mandible
I1	4.5 mo	7.5 mo	4.5y	3.5 y	7.5 y	7.5 y
I2	7.5 mo	10.5 mo	5.5 y	4.5 y	8.5 y	7.5 y
C	7.5 mo	10.5 mo	5.5 y	5.5 y	12.5 y	10.5 y
P1	2.5 y	2.5 y	6.5 y	6.5 y	11.5 y	11.5 y
P2	3.5 y	3.5 y	6.5 y	6.5 y	12.5 y	12.5 y
M1	4.5 mo	4.5 mo	3.5 y	3.5 y	6.5 y	6.5 y
M2	2.5 y	2.5 y	8.5 y	8.5 y	12.5 y	12.5 y
M3	8.5 y	8.5 y	14.5 y	14.5 y	>15.5 y	>15.5 y

at any given time, and all individuals exposed acquire Pb isotope ratios representing those sources; a process that has been named ‘cultural focusing’ (Montgomery et al., 2005). While Pb isotope analysis can be used to identify the movement or migration of peoples with natural Pb exposure, differences in the isotopic ratios from anthropogenic exposure define cultural affinity (Montgomery et al., 2005).

3. Use, misuse, and challenges in Pb analyses

This review specifically examines Pb enamel concentration and isotope ratio studies of archaeological human remains that were previously buried. We found journal articles and book chapters using the search engines: Google Scholar, Web of Science, and OneSearch. Search terms included variations of ‘Pb isotope’, ‘Pb concentration’, and ‘trace element’ with the terms ‘archaeology’, ‘tooth’, and ‘enamel’. Papers were also found in citations from other articles, but dissertations were excluded from the analysis. We examined each article using the same criteria. The article citations and a summarized spreadsheet of the analysis results can be found in supplemental materials (Table S1; SM2).

A total of 55 papers were reviewed (Table S1). They were published between 1979 and 2021. These studies examined populations from the Neolithic to the late-19th-century from all occupied continents. Of these, 15 examined Pb concentration, 11 Pb isotope ratios, and 29 examined both. The following discussion examines the approaches to sample choice, preparation, and the interpretation of Pb concentration and isotope ratios from the papers reviewed. Each section includes a summary of Pb’s use, areas of insufficient information being given or factors influencing conclusions that are overlooked, limitations due to data acquisition or interpretation challenges, and a description of the types of studies needed to help advance this analytical approach.

3.1. Sample selection for Pb measurement in teeth

Researchers have used a variety of types and components of teeth to examine Pb exposure. The choice of an initial sample for analysis is key to avoiding potential contamination or interpretation problems. Most ($n = 36$, 65%) used only the enamel portion of the tooth for the analysis of biogenic Pb or examined the enamel and dentine separately ($n = 13$, 24%). Nevertheless, some studies did use whole teeth or only dentine ($n = 6$, 11%). Due to the potential changes in exposure during an individual’s childhood and the age at which teeth form, it is crucial to describe which teeth are sampled. Chosen teeth included deciduous teeth ($n = 15$, 27%), permanent incisors ($n = 7$, 13%), canines ($n = 17$, 31%), premolars ($n = 23$, 42%), and molars ($n = 39$, 71%). Eleven (20%) did not specify which type of tooth was sampled. A quarter of studies ($n = 14$, 25%) examined multiple types of teeth per individual, usually teeth that form at different ages.

Of those that specified which tooth was sampled, more than half ($n = 30$, 55%) expressed that they systematically chose a specific tooth, but of these, only half ($n = 18$) made a choice based on the age when the tooth developed. The teeth preferred most often were second molars ($n = 17$, 31%) or second premolars ($n = 11$, 20%), which form (depending on the cultural practices) after breastfeeding had ceased (AlQahtani et al., 2010, see Table 1). Others examined teeth from the earliest available developmental period, such as the first molars ($n = 14$, 25%), or, less often, the first premolars ($n = 5$, 9%), incisors ($n = 3$, 5%), and canines ($n = 3$, 5%). Some studies specifically chose third molars ($n = 6$, 11%), which form variably but in later adolescence. Note that the frequencies will not equal 100% since many papers have multiple options of preferred teeth.

There are a few fundamental challenges that arise in sample choice, most of which are common to any isotopic study using teeth. The first is that specific teeth are not always available. This may be due to antemortem or postmortem tooth loss, pathology, taphonomy, previous sampling, or restrictions from the institutions or ancestral groups representing the deceased. Substantial wear or damage can also limit tooth

choice and, in some cases, confidence in positive tooth identification. Since it is common to have small sample sizes and a desire to increase the representation of the population, it is sometimes unavoidable to use non-ideal teeth. However, it is essential that studies report which specific tooth was sampled to link it to the analysis and to consider this for any variations during interpretation.

The preservation of potential sample teeth needs to be considered, though only 18 (33%) specified that they were doing so, and less than half of those ($n = 8$, 44%) defined what that was. Common problems that justify the avoidance of a tooth or areas on a tooth include caries, staining, and cracks because of concerns about contamination after burial (Montgomery et al., 2005; Valentine et al., 2008; Fitch et al., 2012; Shaw et al., 2016). Montgomery (2002) found that poor results of preservation indicators (e.g., surficial staining) did not correlate with the likelihood of the value being diagenetically altered. However, using undermineralized teeth was problematic; this includes using unerupted teeth and enamel close to the cemento-enamel junction.

Various components of chosen teeth were measured in the reviewed papers, including enamel, primary dentine, secondary dentine, and cementum. There is some suggestion that dentine may be more likely affected by diagenetic alteration (Carvalho et al., 2007), however most of this is rhetorical based on Sr studies (e.g., Montgomery et al., 2000). While there are studies that suggest at least some dentine samples may be unaltered (Dudás et al., 2016; Guede et al., 2017), there remains a lack of proper examination of this issue regarding Pb. The foundational studies with Sr were using teeth from basic soils, which are more likely to affect Sr than Pb (Montgomery 2002). How this may translate to acidic soils needs to be examined.

Since 2003, almost all recent studies have exclusively used enamel to interpret *in vivo* values. Not all explicitly stated that specific teeth were avoided, so there may be many researchers who did not avoid or remove areas susceptible to contamination in the burial environment. While it is possible they were taking these precautions, it is necessary to make sure that method descriptions are clear and ideally that methodologies are employed to evaluate their samples for diagenetic alteration (i.e., Kamenov et al., 2018) to minimize possible counterarguments to interpretations and to assist researchers in setting up their own sample protocols.

As part of the initial sample preparation, contamination from cutting and cleaning instruments should be considered. Montgomery (2002) found that the diamond-tipped blades or burrs and agate mortar and pestles were a potential Pb source, instead suggesting to use a tungsten carbide bit to clean all cut surfaces and for the initial surface abrasion or removal of dentine from enamel. While this may be anecdotal and specific to the materials used, this should be considered as a potential source of laboratory contamination that would be difficult to identify. While many studies did not specify the specific materials used for cutting or powdering ($n = 16$, 29%) or cleaning teeth ($n = 11$, 20%), some that used diamond-tipped or unspecified burrs or saws ($n = 13$, 24%) without using tungsten carbide bits to clean after, and others used mortar and pestles for powdering the samples ($n = 4$, 7%). All of these were utilized without specifying whether they had tested their materials for Pb, nor what, if any, laboratory practices or original protocols were in place to ensure that this was not a source of contamination.

Part of minimizing the impact of contamination is considering the expected Pb concentration of the final sample and how this compares to sample loss and instrumentation sensitivity for both Pb concentration and isotope analyses. As of now, instrumentation does not allow for Pb isotope analyses using micro-sampling or laser ablation techniques, therefore all Pb isotope data are from bulk dental samples. While most Pb concentration studies also used bulk samples, laser ablation was sometimes used, some concerns of which are addressed later. The ideal bulk sample mass will vary greatly depending on multiple factors: expected concentration (natural is likely below 1 ppm, and anthropogenic could be much higher, but could still result in <1 ppm concentrations), contamination from pre-treatments and solutions (this will be higher in

isotope pre-treatments since there is interaction with more solutions), expected loss of concentration during pre-treatment (either from taking aliquots for different analyses from a single bulk digest, or from the column chemistry to elute Pb for isotopic analyses), the target voltage intensity for analysis (ensuring it is at least an order of magnitude about the limit of detection and quantification), the final solution concentration, and if applicable, volume, needed to reach that target intensity. While Dudás et al. (2016) did examine what the low limit of sample size mass should be, it was unclear if heterogeneity below 10 mg was from an unknown preparation contaminant or *in vivo* variation. When specific methods were stated in the reviewed papers, it was often reported that 10 mg was used for concentration and a further 50 mg for Pb isotope analyses; however, most gave no or insufficient information regarding sample mass. While there is no universal ideal sample mass, future studies should be more detailed in terms of the specific steps taken to avoid contamination.

While consideration of avoiding diagenetic materials and post-depositional contamination is important to data quality, there is a current challenge in whether to use a bulk or micro sampling technique when acquiring Pb concentrations. Laser ablation has been utilized in method development to examine Pb concentration patterns within teeth (e.g., Budd et al., 1998; Webb et al., 2005), but more recently, studies have linked changes in concentration to the age of formation (Jones 2015; Quinn et al., 2020). However, there is ongoing debate as to whether the offset in the relationship between the timing of tooth formation and enamel maturation is sufficient to undermine the usefulness of serial sampling connected to physiological age (e.g., Smith and Tafforeau 2008; Montgomery et al., 2010; Simmons et al., 2013). When enamel forms, the initial composition is rhythmic, creating cross striations and striae of Retzuis, which can be used to estimate the formation time (Lacruz et al., 2017; Berkovitz et al., 2017). The mineralization of the resulting enamel happens in two phases: secretion and mineralization. The initial secretion is more organic and is replaced with highly mineralized hydroxyapatite, which matures from the enamel-dentine junction towards the surface (Al-Mosawi et al., 2018). There is a slight offset in timing between secretion and mineralization, unlike in dentine. However, the specific pattern is still unclear and may be affected by overprinting or averaging of the elemental/isotopic history of the enamel. While there is promising evidence to suggest that the effect of offset and overprinting is minimal (Austin et al., 2016; Green et al., 2018; Smith et al., 2018; Müller et al., 2019; Dean et al., 2020), and that the Pb present in the final mineralized material was from the initial secretion (Müller et al., 2019), the amount of time represented is still unclear, particularly due to the long residence time of Pb in the body compared to other elements examined in the studies. Though there is less temporal specificity, a more certain method to examine change over time within an individual is by using bulk samples from multiple teeth (i.e., Valentine et al., 2015).

While many studies systematically choose which teeth to sample, the lack of reported details in the sample choice procedure creates two challenges for data interpretation: the potential remobilization of Pb from past exposure and the impact of social age on interpretation. Deciduous teeth form *in utero* and early infancy, while adult teeth form after birth (AlQahtani et al., 2010). Pb concentrations in first molars, or other teeth that form at a similar time (see Table 1), provide a record of the period closest to the time of birth when infants and children are most susceptible to Pb through hand to mouth activities and via their underdeveloped digestive system. However, remobilized maternal Pb can impact teeth that form early in development or during breastfeeding (Gulson et al., 2003). Any exposure from the physical environment to infants could be mixed with maternally transferred Pb. In situations where the mother has moved to an area with different bioavailable Pb isotope ratios, the Pb exposed to the developing fetus/infant would be mixed with, if not entirely swamped by, non-local values (Gulson et al., 2003). This challenge is compounded when the study does not identify the specific tooth sampled for each individual, thus making it difficult

for readers to assess if this should have been a potential source considered by the researchers, given the age at development.

Understanding specifics about the type of tooth sampled is vital because of both physiological and social differences in the potential of Pb exposure and its pathways. There are (at least) three types of age: chronological, biological (physiological), and social (Kamp 2001:3; Baxter 2005:19; Halcrow and Tayles 2011:335). Chronological age is the time elapsed since birth, either in days, months, or years. Biological age considers the biological changes that occur to an individual, such as physical growth or the development and eruption of dentition – these developments demarcate biological age groupings. Social age groups are boundaries of normative behaviour and are based on culturally constructed views of maturation. Bioarchaeologists use biological age data, which they infer as chronological data, then look for patterns to determine the chronological boundaries of social age (Sofaer 2011:290). However, these boundaries can be challenging to conceptualize, particularly for societies and periods with little to no written records regarding children (Sofaer 2011; Agarwal and Glencross 2011). As Halcrow and Tayles (2011:251–252) suggested, there is no easy answer to this problem, and more work needs to be done with theoretical views of the body.

Studies must approach skeletal remains while considering that these individuals, even as children, were active members of society who experienced numerous physical and social changes. In turn, differences in exposure between teeth may be due to social differences, not just changes in geographical environments. Without a systematic sample choice that includes the consideration of social factors during childhood, the original researchers' data could be misinterpreted and does not allow those examining childhood in a social setting to re-examine the work in the future. Although not all bioarchaeologists need to adopt the childhood social theory framework, these issues need to be considered. Until such a reconciliation can be made, what is most important is to be clear about the methods used, the terminology employed, and to integrate historical and archaeological context whenever possible.

3.2. Interpretation of Pb concentrations over time

Pb concentration of tooth enamel in contemporary populations is a valuable tool to track exposure during tooth formation (e.g., Gulson 1996; Robbins et al., 2010). These data were used for several purposes, including method development or refinement, examining change in concentration over time, comparing the general pattern between populations, inferring health effects, and examining diagenesis. Some studies did not interpret the values in the papers ($n = 4$, 10%), either because the purpose was method development (Budd et al., 1998) or other stable isotope systems were more effective at identifying outliers or dietary patterns within the population (Szostek and Glab, 2001; Müller et al., 2003; Keller et al., 2016).

Studies were inconsistent in the amount of information given regarding concentration methods, and less than half gave sufficient analytical methodology information for the Pb concentration results to be replicable ($n = 19$, 43%). Many lacked basic information on the initial dilutions, instrumentation, standards, error (of standards and individual samples), and calibration corrections. Earlier research used atomic absorption spectroscopy (AAS) to determine concentrations, but since 2000 it is more common to use hexapole or quadrupole inductively coupled plasma mass spectrometry (Q ICP-MS) or isotope dilution thermal ionization mass spectrometry (ID-TIMS) (Table S1). Rather than analyze cross sections of enamel using laser ablation ($n = 5$, 11%), most digested the samples in acid prior to analysis ($n = 35$, 80%), and of these, almost all utilized the bulk digest solution ($n = 30$, 86%). There were four studies (10%) that determined concentration after column chemistry and some without the use of an isotopic tracer (Table S1). However, column chemistry techniques are rarely 100% efficient, so this can miscalculate the Pb concentration and is therefore not advisable.

For studies that interpreted the concentration values, there were

differences in what the samples were compared to and how the data were converted for interpretation. The four main approaches that were used for interpreting data were: 1) to consider the general pattern of Pb exposure, often comparing it to previous modern and/or archaeological studies; 2) to map changes in Pb exposure in teeth over time to understand high modern exposure; 3) to convert enamel to blood Pb or comparing the data to modern values to infer health; and 4) to compare trace element concentrations to previous studies, or isotopic ratios of the soil, to suggest diagenetic alteration (Table S1).

The most common approach was to compare values in archaeological samples to either modern ($n = 8$, 18%) or archaeological ($n = 16$, 36%) teeth from similar contexts, or both ($n = 11$, 25%) (e.g., Montgomery et al., 2005; Price et al., 2017; Table S1). Unfortunately, there are limited data for directly comparable modern values of Pb in teeth. Modern studies typically use surficial enamel, cervical portions of teeth, or whole teeth to produce their data (e.g., Fosse et al., 1995; Gulson 1996; Gomes et al., 2004), while most archaeological studies sampled core enamel. This creates a problem for cross-comparison since the concentration of Pb is not uniform throughout the tooth (Budd et al., 1998). The surficial enamel is highly enriched compared to core enamel or dentine, and cervical portions of teeth include both dentine and enamel, and since dentine has higher concentrations of Pb, comparisons will result in erroneous interpretation of differences. Comparisons of concentrations taken from different portions should not be employed. Therefore, comparisons of concentrations taken from different portions should not be employed.

In comparing archaeological populations, part of the consideration is whether the exposure was from natural or anthropogenic Pb sources. Since natural levels of Pb exposure result in low enamel concentrations, high enamel concentrations help to identify when anthropogenic Pb exposure occurred (Montgomery et al., 2005). Several studies established that concentrations below 0.5–0.8 $\mu\text{g/g}$ (ppm) can be considered exclusively natural Pb exposure (Montgomery et al., 2010; Millard et al., 2014). However, Pb concentrations can vary between individuals within populations even when the overall population exposure is exclusively high or low (Budd et al., 2004). Schroeder et al. (2013) used this concept of very low natural Pb exposure to distinguish between enslaved labourers at the Newton Plantation who were born locally in Barbados and those born in Africa. The African environment was inferred to have limited atmospheric exposure and no Pb-containing material culture. These concentrations were further extrapolated to examine health impacts on those working at the plantation. Although there is a trend in bioarchaeology to examine life histories of individuals (Katzenberg 2008), limitations in methods converting enamel to blood Pb concentration must be considered.

Some of the interest in using Pb concentrations in teeth is to examine the potential health effects of past exposures. A fundamental challenge with interpreting Pb concentration data is the uncertainty in converting Pb enamel concentrations to blood Pb. While there is considerable information on blood Pb levels and their association with adverse health impairments (e.g., Needleman 2004), the extrapolation of using tooth enamel Pb concentration to determine the extent of impacts on health is poorly understood. To examine these health impacts, several studies (Montgomery et al., 2010; Schroeder et al., 2013; Quinn et al., 2020; Moore et al., 2021) converted Pb concentration in enamel to blood Pb using a 10:1 blood-to-enamel ratio proposed by Grobler et al. (2000). Grobler et al. (2000) state that the conversion they developed should not be used with high enamel concentrations or in situations where people are moving between regions. Furthermore, the 10:1 ratio proposed ignores the variability of both blood and enamel data from Grobler et al.'s (2000) initial study. The coefficient of variation was greater than 70% of the mean (enamel: 0.33 ± 0.27 mg/kg and blood: 0.031 ± 0.022 mg/kg). Therefore, the ratio could actually vary between 1:1.3 and 67:1, which is too large a difference to be applicable without additional study. Finally, the teeth used in the study by Grobler et al. (2000) formed *in utero*, yet the blood concentrations were taken from children at 6–8

years old. Therefore, there may be discrepancies in the amount of Pb to which a fetus versus a child were exposed. Unfortunately, researchers have not acknowledged these limitations, and while natural versus anthropogenic exposure can be interpreted at the individual level, the absence of a simple conversion method means it is only possible to interpret potential health effects at the population level.

At the population level, it is possible to compare between sites. A meta-analysis comparing modern and archaeological tooth concentrations would be beneficial to understand the change of exposure over time and to allow for better inferences regarding past health. While it may not be possible to develop an accurate equation to correlate enamel concentrations to blood values to make inferences about health, it is useful to explore this possibility. Additionally, while we now have a good estimation of what the natural Pb exposure was in archaeological populations of Britain (Montgomery et al., 2010; Millard et al., 2014), this has not been examined in other geographic areas. As such, studies are needed on the concentrations of Pb in archaeological populations from more geographically and culturally diverse settings to better delineate the cutoff between natural and anthropogenic exposure.

3.3. Interpretation of potential Pb sources

Pb isotope ratios are used in the archaeological examination of populations in several ways, including diachronic change over time, geographic movement, and diagenesis. While early research compared modern and archaeological exposure to see change over time and the relationship to atmospheric pollution (Patterson et al., 1991; Budd et al. 2000a, 2004), Montgomery et al. (2000) were the first to apply Pb isotope ratios in teeth to examine human geographic movement. Fundamental to addressing these topics is the acquisition of high-quality data.

Pb isotope data are acquired using a few different approaches but starts with column chemistry to isolate the Pb. Two main forms of resins are used: anion resin (i.e., AG-1X8 or Dowex 1 $\times$ 8; $n = 23$, 58%) or a crown ether resin (i.e., Sr-Spec; $n = 6$, 12%). The column chemistry required to elute Pb from these resins varies, but regardless has the basic steps of cleaning, conditioning, loading sample, washing, and elution – with specific solution volumes. In the last few years many authors do give sufficient information, either in the paper or through citations, to replicate their method ($n = 10$, 25% total and $n = 7$, 18%, partially), but some still include broad descriptors such as ‘HBr ion exchange’ ($n = 7$, 18%) or ‘anion exchange column chemistry’ ($n = 5$, 13%). Beherec et al. (2016) suggested that Sr-spec resin results in too high of blanks, but this may depend on the column chemistry involved, which was not stated. The impact of the blanks would also depend on what the expected concentration of the sample was and the initial sample mass. Blanks have not been a problem for the other studies that used this technique, and the resin allows for Sr to be eluted before Pb (Pin et al., 2014) during the same column chemistry. It is possible to collect the washes from the anion exchange chemistry and use it to load onto a separate Sr specific resin column after, but this increases cost and time.

With Pb eluted, the solution is analyzed using MC-ICP-MS ($n = 23$, 58%), or dried onto a filament and analyzed through TIMS ($n = 14$, 35%). While the specific analytical parameters will depend on the analytical machinery available, certain information remains important to state. Parameters include the analytical mode (dynamic/state), cones used, faraday array, the timing of baseline and integration along with the number of blocks of data. Furthermore, there remains little explanation of calibration methods. Though many do use thallium (Tl) corrections with a thallium spike into all samples and SRM 981 as a standard ($n = 23$, 58%, $n = 11$, 28% not specified) to account for mass bias, Kamenov et al. (2004) found that there is significant variation in the $^{205}\text{Tl}/^{203}\text{Tl}$ ratio, so the time elapsed since solution mixing needs to be controlled and accounted for, but is not always reported in these studies. Data can also be calibrated using the most probable values of standards (i.e., SRM 981, $n = 6$, 15%) bracketed after a certain number

of samples, then using an average or linear equation to correct for drift that occurred between data brackets. It should be noted that there are some different Pb isotope values of SRM 981 that data can be normalized to, but few authors were clear about what values used were or citations that they originated from. All these parameters and calibration techniques have impacts on data quality, but there remains a need to examine, with interlaboratory studies, how big of an impact such facts cause for archaeological data. It was slightly more common in Pb isotope analyses than with concentration that analytical methods were clearly stated ($n = 21$, 48%) or at least partially so ($n = 14$, 35%). However, when sample procedure, pre-treatment, and analytical sample methodology are all considered only three studies truly gave enough information for their Pb isotope analyses to be reproducible: Turner et al. (2009), Millard et al. (2014), and King et al. (2021).

Related to standards, is the reporting of data quality. To assess the quality of data, both percent error to understand accuracy and variability to understand repeatability need to be reported. However, only one (3%) study gave information on both accuracy and repeatability, many provided only accuracy ($n = 15$, 41%) or had insufficient context to interpret ($n = 8$, 22%). Those that gave information on variability ($n = 16$, 43%) in all but three instances (81%) gave sufficient information to back calculate the accuracy, even though not directly addressed. Since absolute differences in Pb isotope ratios of standards to known values and the 2σ of the analyses are often in the 2nd–3rd decimal for $^{206}\text{Pb}/^{204}\text{Pb}$ and 3rd–4th decimal for $^{207}\text{Pb}/^{206}\text{Pb}$, and meaningful differences are often close to this level of error, high quality data is necessary. The propagated error (2σ) of each individual sample should also be given, which will account for both the variation in the individual analysis and that added from the calibration. It should also be noted that reporting on data used for calibration as a means of addressing quality is not appropriate, so additional analyses of SRM 981 during a run, or a different check standard, is necessary. For example, SRM 1400 (Bone Ash) is a bioapatite standard with most probable values published in Hinniers et al. (1998). Since it is processed through the same column chemistry procedure as the tooth samples it is matrix matched and can account for any such problems introduced during analysis. Either way, researchers should be clear about the errors of the samples.

Migration studies remain the primary application of Pb isotope data, but within these, there are three approaches. The first examines populations that are determined to be free of potential anthropogenic Pb exposure and identifies non-locals when their Pb isotope ratios do not fall within the expected local isotopic range (e.g., Budd et al., 2000a; Dudás et al., 2016; Samuelsen and Potra 2020). The second approach infers a cultural focusing of Pb isotope ratios with an increase in Pb concentrations and identifies non-locals to a cultural region (e.g., Budd et al., 2004; Montgomery et al., 2005). The third combines the first two but uses the $^{238}\text{U}/^{204}\text{Pb}$ ratio, which aids in separating between natural (silicate) and anthropogenic (sulphide) Pb sources (Evans et al. 2018 a; b). While movement studies mainly focus on the population level, researchers are increasingly examining the life histories of individuals (e.g., Budd et al., 2000a; Valentine et al., 2008; Lamb et al., 2014; Laffoon et al., 2020). A related but separate use of Pb, is for the examination of diagenetic alteration of tooth enamel by considering Pb isotope ratios and other trace elements as markers of potential influences on data interpretation (e.g., Dudás et al., 2016; King et al., 2020; Simonetti et al., 2021).

To infer movement, the locally available Pb source must be characterized. There are a few general approaches, though many studies used more than one approach (Table S1). The first is using existing or new data regarding the local environment ($n = 28$, 70%). An alternative is to utilize existing publications that characterized the Pb ore known to be available locally ($n = 19$, 49%). Finally, some researchers use ranges from previously published archaeological human comparative data ($n = 20$, 45%). The approach taken depends on whether it is determined that the population was exposed to anthropogenic or natural Pb. In the case of natural exposure, it is not simply the local geology, but the

bioavailable Pb that requires isotopic characterization. However, of the 28 studies that examined natural exposure, this concept was mentioned or considered by only 19 (68%) and defined by 13 (46%).

There is no universal method to define the bioavailable Pb in natural exposure contexts using the local environment. Many looked at the underlying geology ($n = 15$, 54%), but no studies used only this method. It was also common to sample the burial soil matrix or local subsoil ($n = 14$, 50%), and archaeological bone or teeth from fauna ($n = 8$, 29%). Less often, the mean and standard deviation of bone, dentine, or enamel Pb isotope ratios from the population itself were used ($n = 5$, 18%).

While numerous reviews (e.g., Price et al., 2004; Bataille et al., 2020; Holt et al., 2021) and applications define bioavailable Sr ratios at the local or regional level (e.g.; Montgomery et al., 2010; Willmes et al., 2018; Janzen et al., 2020; Wang and Tang 2020; Frank et al., 2021; Washburn et al., 2021), there is limited discussion on how to apply Pb for this purpose (Giovas et al., 2016; Sharpe et al. 2016, 2022; Tschetsch et al., 2020; Samuelsen and Potra 2020; Price et al., 2021). While for Sr it is ideal to sample flora, fauna, and or water samples to characterize the bioavailable values (Bentley 2006; Evans et al., 2010), contamination by the widespread use of anthropogenic Pb since industrialization complicates this process with Pb (Samuelsen and Potra 2020). Therefore, context-specific archaeological fauna must be sampled.

While several studies have followed this approach (i.e., Valentine et al., 2008, 2015; Turner et al., 2009; Turner and Armelagos, 2012; Price et al., 2017, 2019; Evans et al., 2018a; Samuelsen and Potra 2020), there is limited discussion in the discipline of what animals (wild or domesticated) are acceptable for such studies and there is not enough attention to characterizing the variability to be able to define an adequate sample size to detect change. For example, Sharpe et al. (2022) has suggested that decisions of animals will depend on the availability and likelihood of local geology to not diagenetically alter the materials, but that thicker enamel may fare better. Many studies use soil/sediment samples ($n = 14$), but it is questionable whether they are sufficient to characterize local bioavailable Pb, particularly given the potential of anthropogenic soil contamination. This also assumes that the soil in which people are buried represents the land where they hunted, gathered, and or grew their food, which is certainly not always the case. While variability will depend on the complexity of the geology, foraging or agricultural practices, etc., statistical power warrants further focus. Sample choice is often dictated by what is available to the researchers, but the limitations of this approach need to be articulated and considered during the interpretation of results.

Studies that attributed exposure to anthropogenic sources ($n = 24$, 60%) often examined sources in their cultural environments because of the cultural focusing of Pb. Studies have attempted to characterize the cultural environment using either Pb isotope ratios at the mines that were known or suspected to be the primary sources of Pb to that population (e.g., Fitch et al., 2012), or used atmospheric deposits to establish expected bioavailable values (Fitch et al., 2012; Keller et al., 2016; Aronsen et al., 2019; Eshel et al., 2020). Budd et al. (2004) established that there is no direct connection between human Pb isotope ratios and those in the physical environment (i.e., soil cores and other atmospheric deposits), so the applicability of atmospherically derived baselines is unclear.

Defining bioavailable sources in populations exposed to anthropogenic Pb presents a different challenge. Part of this stems from the assumption that exposure patterns in modern contexts hold true in the past. Kamenov (2008) argued that Pb in teeth from modern Bulgarian individuals is mainly from dust associated with tetraethyl Pb in gasoline that is inhaled or ingested, which is an observation supported by studies in other modern populations (Mielke and Reagan 1998; Zhang et al., 1998; Gwiazda et al., 2005; Laidlaw et al., 2005). However, unlike many populations in the past, none of these modern populations frequently used material objects containing high amounts of Pb. Therefore, exposure to these materials likely contributes significantly to the body's Pb burden. In turn, the idea that Pb comes mostly from inhalation rather

than ingestion cannot be indiscriminately employed in past populations. Yet, this concept is occasionally applied both in natural (Turner et al., 2009; Jones et al., 2017; Price et al., 2017; Scaffidi et al., 2021; Turner, 2021) and anthropogenic exposure contexts (Beherec et al., 2016; Keller et al., 2016; Price et al., 2019), sometimes leading authors to limit their consideration of other Pb-containing materials.

Rather than assume that sources from the physical environment are the likely contributor, the material cultural environment must be explicitly considered. Although, even in modern contexts, mines are not often the exclusive Pb source to a population (Gulson 2008), it is common for studies to consider only the locally mined Pb (e.g., Fitch et al., 2012). Presumably, this is assuming that the mined Pb would pollute the environment they are living in and cause exposure. Based on Mahaffey (1978), Montgomery et al. (2010:219) suggested that there were three aspects to exposure from Pb: “1) Drinking soft water or acidic fluid from Pb pipes, cisterns or vessels; 2) Ingestion of bioavailable Pb; or; 3) accidental ingestion of Pb from glazed pottery.” Ingestion or inhalation of dust would fall into the second category, yet many studies ignore the others. Arguably, we can add a fourth possibility of dermal Pb exposure through body paint or makeup. These potential sources could easily contribute to the overall exposure of a population or even mask the role of the ingested or inhaled bioavailable Pb from the physical environment. Since Pb used in those materials may not match any pollution to the local environment, all sources of Pb to a population must be considered.

Attributing Pb to anthropogenic or natural exposure involves considering a population's sources of Pb, both from the environment and the potential material culture that was locally sourced and produced or imported. While it is impossible to map all potential sources, there is information in archaeological materials, ethnographic accounts, and historical records that help suggest the specific materials. Following this, consideration of the Pb source within the materials should be examined. Part of the anthropogenic exposure can be through material goods, even imported goods. Although some studies do mention the potential Pb exposure to such materials, it is largely discounted or ignored entirely (e.g., Fitch et al., 2012; Keller et al., 2016; Aronsen et al., 2019; Quinn et al., 2020), or the characterization of all Pb sources to a population is ignored (e.g., King et al., 2020).

An additional limitation in the approach to Pb is in regard to what populations should be considered to have ‘natural’ rather than ‘anthropogenic’ Pb exposure. This can affect both when exposure is attributed to only natural or only anthropogenic sources. Referencing Turner et al. (2009), Kamenov and Gulson (2014) present a version of this critique by asserting that the Inca did have access to Pb-containing materials. However, this has gone unheeded by some recent studies and remains a potential alternative interpretation rather than an indication of diagenetic alteration (i.e., Turner et al., 2009; Scaffidi et al., 2021; Simonetti et al., 2021). Considering possible Pb sources outside of populations typically thought to have metallurgy is important since researchers may be overlooking some anthropogenic Pb sources. For example, Jones et al. (2017) found that some individuals at a site in the Lower Illinois River Valley during the Middle Woodland Period (2100–1700 BP) had Pb isotope ratios similar to locally available galena and proposed that it may be associated with later ethnographic accounts of its use as a ceremonial pigment. This possibility is compelling and highlights the potential of ethnographic and archaeological records to support alternative conclusions.

In cases where it is unclear whether exposure occurred exclusively from anthropogenic sources, it is advisable to not only examine the Pb concentration of the enamel, but also characterize the local Pb environment, ideally using faunal samples (the number of which determined through power analyses). This is in addition to considering the Pb isotope values of anthropogenic sources. If the anthropogenic values are too similar to the local environment, attribution to anthropogenic exposure may not be possible, however incorporating $^{238}\text{U}/^{204}\text{Pb}$ ratio (Evans et al., 2018a, 2018b) may help differentiate between natural and

anthropogenic Pb sources. Focusing on the concept of ‘natural’ exposure was effective in the 1980s to early 2000s in establishing high levels of contemporary Pb. However, it is not always clear what populations had solely ‘natural’ exposure levels. Although there are cases where interaction with anthropogenic Pb occurred, that is not always the case. Studies should consider the cultural environment and materials people may have interacted with when determining potential Pb exposure sources.

Rather than assume that populations were exposed to solely ‘natural’ Pb from the underlying geology, archaeological and historical evidence along with Pb concentrations should be considered to critically assess the point. Whether natural exposure is concluded or it is determined that anthropogenic exposure occurred, this should remain contextualized in the study using the archaeological and historical data by examining what Pb geological sources were common, and what objects could they have been using (both locally produced and imported).

The final concern about interpreting potential movement and sources of Pb relates to how diagenesis might alter Pb concentrations and isotope ratios. Of the studies reviewed 46 (84%) considered diagenesis. Of those that concluded diagenesis possibly or did occur ($n = 18$, 33%), three methods were used to come to this conclusion (Table S1). First, was when elemental concentrations or ratios were considered to be high or were high in comparison to specific cut-offs established by Patterson et al. (1991) or Kamenov et al. (2018) ($n = 8$, 44%). Next, if the Pb concentration was higher than expected in cases of natural exposure or they were too high to be biologically possible ($n = 5$, 28%). Finally, when Pb isotope ratios were outside of the local or expected range or suggest an additional source through a linear array/mixing model ($n = 3$, 17%). Of those that concluded diagenesis occurred, only five (28%) specified that they were considering the preservation of the individual teeth, and one defined what characteristics were considered in establishing the level of preservation. However, half of the examined studies did consider but did not identify that diagenesis occurred ($n = 28$, 51%). While many of these did not examine any criteria ($n = 17$, 37%), instead saying enamel was stable ($n = 12$), the same techniques used by those who determined diagenesis occurred were applied. Therefore, not finding diagenesis was not exclusively an issue of papers not having examined the issue.

Considering the methods applied to identify diagenetic alteration, studies may be misattributing the population's use of Pb or the results from poor sample choice and preparation to the effects of diagenesis. Additionally, establishing arbitrary cutoffs for what is expected for ‘normal/natural’ Pb concentrations in teeth is not advisable without solid consideration of data error and reliability. Finally, there are no direct quality checks to determine if diagenesis has occurred beyond examining if other trace elements are above what is expected. Kamenov et al. (2018) established parameters of what should be considered normal using specific trace element concentrations; however, what is considered within the ‘normal’ range is based on modern enamel studies and a small data size from Patterson et al. (1991). Moving forward, more consideration of sample choice and analytical methodology should be considered when interpreting diagenesis.

4. Conclusion

In bioarchaeology, Pb analyses are currently used to address differences in Pb exposure between populations, identify instances of natural vs anthropogenic exposure, and reference overarching health concerns. Pb isotope analysis is applied to examine the movement of people in cases of natural exposure and movement out of a broader region if anthropogenic exposure occurred. Pb is a powerful analytical tool, particularly when used with other isotopic systems. However, a few underlying themes of misuse need to be considered, that are summarized in Table 2. First, the assumption of anthropogenic versus natural exposure must be critically evaluated using archaeological, ethnographic, and historical documentation. Only with this information can we more

Table 2

Summary of topics regarding use or application, misuse or insufficient information given, challenges from current limitations, and studies needed within each theme explored as part of the review.

Issue	Use	Insufficient information or consideration	Current limitation	Study Needed
Sample choice	- Mostly enamel - Most frequently premolars or molars - Some systematically for age - Bulk or micro sampling	- Preservation - Contamination by cleaning/cutting - Unidentified teeth - Remobilization of Pb during pregnancy and breastfeeding	- Tooth availability - Tooth enamel maturation timing - Biological versus cultural age	- Dentine diagenetic alteration
Interpretation: Concentration	- Method development - Comparing general population patterns - Inferring health effects - Examining diagenesis	- Insufficient method information - Converting enamel Pb to blood Pb - Concentration not always examined in “natural” exposure contexts	- No directly comparable modern values	- Meta-analysis of modern and archaeological tooth values - ‘Natural’ Pb exposure in other regions
Interpretation: Isotopic ratios	- Movement by identifying individuals outside local bioavailable range or outside range for locally used ore - Examine diagenetic alteration	- Insufficient method information - Not considering both atmospheric and cultural material exposure - Many Indigenous populations assumed to have no anthropogenic exposure	- No standard for defining bioavailable Pb - Identifying all Pb sources to a population - Identification of diagenetically altered samples	- Best practices for establishing bioavailable values

holistically consider the sources of Pb, including the materials that result in Pb exposure. Additionally, studies give insufficient information on sample choice, preparation, and data acquisition, which means that the methods are not reproducible, thus negatively affecting the interpretation of the data. Moreover, the readers cannot assess if methodological, physiological, or cultural limitations have been considered, which weakens the studies’ conclusions. Finally, we do not have a means to directly identify if Pb diagenesis has occurred. The most common method is via proxies of other elements that do not necessarily have the same reactions in the burial environment. This is compounded by insufficient reporting of sample preparation and analytical methodology. Without properly defining potential Pb sources to a population and potential sampling or analytical methods defined, the arguments questioning the stability of enamel bioapatite are greatly weakened.

Based on this review of 55 papers dealing with Pb concentration and isotope analyses in bioarchaeological contexts, we make the following recommendations. First, studies must specify the specific tooth sampled for each individual. This is important because of remobilized Pb during breastfeeding and the potential for cultural influences that can affect how an individual may be exposed to Pb over the course of their childhood. While laser ablation is used, it is still uncertain if changes in Pb exposure relate to a specific age of development due to the unclear zonation in teeth. Therefore, more Pb studies may be necessary before we can assess the specific health experiences of an individual at a yearly level. Though it cannot specify health impacts to an individual, determining Pb concentration is important to assess differences in Pb exposure between populations and for identifying an individual’s potential for anthropogenic exposure. However, many studies examining ‘natural’ Pb are not acquiring concentration data, which would be an important additional means of supporting the assumption that there was no anthropogenic exposure to a study population.

Finally, there need to be more studies on characterizing bioavailable Pb isotope ratios in anthropogenic and natural contexts. Pb from material goods is often underemphasized or ignored in anthropogenic Pb source interpretation. These can be a substantial or the only source of Pb to a population. Yet, there remains a focus on just the atmospheric Pb or ore and a disregard for the Pb-containing objects with which populations would have been interacting. While [Samuelsen and Potra \(2020\)](#) asserted that animal teeth are ideal for characterizing bioavailable Pb isotope values in natural contexts, the number of samples necessary will depend on data variability and should be explicitly examined using power analyses.

It is hoped that the guidance and recommendations provided in this review will increase confidence in results and improve the ability to interpret specifics about Pb data reported, to compare studies, and to transfer protocols. Appropriate use of the techniques requires careful

consideration of sample choice, determination of possible contamination, and data interpretation would benefit from improving the contextual information available from archaeological, ethnographic, and historical documentation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Image based figure made using <https://biorender.com/>.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jas.2023.105803>.

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